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Exploiting the Binding-Conversion Disparity in Prion Propagation: A D-Amino Acid Approach to PrP^{Sc} Sequestration

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Introduction

Prions are proteinaceous infectious agents that cause neurodegenerative diseases that can arise through inherited, acquired, or sporadic mechanisms [1]. In particular, the conversion of the cellular prion protein, from the normal form PrP^{C} to the pathogenic isoform PrP^{Sc} is involved in the development of transmissible spongiform encephalopathies [2]. PrP^{Sc} is resistant to conventional sterilisation methods and is also able to withstand environmental extremes due to their protein-only structure and high stability [3].

The pathophysiology of human prion disease is partially understood, with the prion hypothesis well-supported but the downstream effects incompletely characterised. Identified contributors to prion disease neurodegeneration include activation of the unfolded protein response and dysfunction of the ubiquitin-proteasome system [4]. Critically, propagation and accumulation of PrP^{Sc} is central to disease progression and associated with the neurotoxicity characteristics of prion disease [5].

Thus far, no effective cure or treatment for human prion disease has been discovered, although multiple therapeutic strategies are under investigation, with some having entered clinical trials. The main concepts for these therapeutics are 1. Reducing conversion of PrP^{C} 2. Preventing propagation of PrP^{Sc} 3. Alleviating downstream effects, e.g., neuroinflammation. 4. Clearance of existing PrP^{Sc}. Challenges facing the discovery and/or development of effective treatments include non-specific interactions, toxicity, and drug resistance [6].

The study of prion disease is important not only due to their severe and currently invariably fatal clinical outcomes, with a fatality rate of 100% once symptoms arise, but also because of their broader biological and public health implications. The potential for prions to be used in the context of biological weaponry underscores the need for effective therapeutic strategies [7]. There is also a risk of zoonotic spillover from animal-to-human transmission, as shown with the precedent of variant creultzfeld-jakob disease, and particularly of concern currently is the virulent chronic wasting disease that devastates cervid populations, although the risk of development of cross-species transmission abilities is estimated to be low [8, 9].

Research questions and objectives

This research proposal aims to explore a therapeutic strategy for interrupting propagation of PrP^{Sc} by exploiting a potential disparity between the molecular requirements for PrP^{C}-PrP^{Sc} binding and those required for conversion. D-amino acids are the chiral enantiomers (mirror-image stereoisomers) of L-amino acids, which constitute the predominant biological form utilized in protein synthesis. Specifically, this study seeks to address two research questions:

1. Can a D-amino acid construct mimicking key PrP^{C} binding regions and their native spatial arrangement bind to PrP^{Sc} without conversion, with an affinity comparable to the native PrP^{C}-PrP^{Sc} interaction?
2. What structural features of the construct (e.g., crosslinker rigidity, spacer length) would allow for the long-term persistence of the resulting complex against PrP^{Sc} dissociation and continued propagation?

For the purposes of this study, a stable dead-end complex is defined as the structure formed through interaction between the D-amino acid construct and PrP^{Sc} that has inactivated the bound PrP^{Sc} and prevented it from participating in further templated conversion.

Literature Review

PrP^{C} is an α-helix rich membrane protein found in the central nervous system [10], while PrP^{Sc} is high in β-sheets and often assembles with other PrP^{Sc} molecules to form aggregates [11]. During seeded amyloid propagation, PrP^{Sc} converts PrP^{C} into the misfolded isoform, and elongates the amyloid fibrils with newly converted PrP^{Sc} incorporated into the growing end. The elongated fibrils then undergo fragmentation, increasing the number of independent aggregates available for further propagation [12]. The conversion of PrP^{C} to PrP^{Sc} is thought to involve binding of incoming PrP^{C} to the ends of PrP^{Sc} fibrils, which is then followed by extensive conformational restructuring and incorporation into the growing end [13]. The disordered N-terminal region of PrP^{C} is suggested to interact with the PrP^{Sc} templating surface through hydrogen bonding between complementary –NH and –CO groups, inducing a cross-β conformation to be adopted in the region [14]. The remaining folded domain subsequently undergoes extensive unfolding before being refolded into the PrP^{Sc} structure [13, 14].

As mentioned in the introduction, there are multiple categories to prion therapeutics. Preventing or reducing conversion of PrP^{C}, either through protecting PrP^{C} from conversion or reducing the numbers of available PrP^{C} [15], often only intercepts before recruitment and would not act on the PrP^{Sc} that have already been established. This is particularly of concern as prion diseases are often diagnosed in the later stages due to similarities with other neurodegenerative diseases [5]. Alleviating the downstream effects of prion disease may aid in slowing cognitive deterioration but is unlikely to stop the spread of PrP^{Sc} itself. Reducing the levels of PrP^{Sc} is the ideal treatment for prion disease as it directly addresses the problem, however it faces different challenges, such as the β-strand-rich, multimeric architecture of PrPSc fibrils, which confers protease resistance and shields individual monomers from degradation [16]. Thus, the prevention of propagation, acting on the templating interface on PrP^{Sc} to prevent the recruitment of more substrate, has been chosen as it acts on the already-existing PrP^{Sc} pool, without requiring the breakdown of the protease-resistant aggregate.

The requirements for binding and conversion are not likely to be synonymous, with PrP^{C} mutants still able to bind to the PrP^{Sc} template source without conversion [17]. The relatively 'promiscuous' binding of PrP^{C} with PrP^{Sc} template due to the need for only a few regions on the PrP^{C} molecule, however, conversion has high sequence selectivity due to the need for compatibility between the primary structures of the PrP^{C} and PrP^{Sc} molecules [18]. As productive prion conversion requires substantial conformational restructuring of PrP^{C}, stereochemical compatibility may be critical for conversion [19]. However, whether the same stereochemical requirements apply to the initial PrP^{C} – PrP^{Sc} binding interaction remains unclear. It suggests a window in which a molecule satisfies the geometric/chemical requirements of binding without ever satisfying the stricter, sequence- and stereochemistry-dependent requirements of conversion. An all-D-enantiomeric peptide has been shown to compete with PrP in the binding to amyloid beta (Aβ) oligomers. This competition disrupted the heteroassembly of Aβ-PrP. This implies that stereochemically inverted peptides could still retain sufficient molecular recognition to interfere with protein-amyloid interactions [20]. Translating this strategy to the context of PrP^{Sc}, we propose that a D-amino acid construct mimicking key PrP^{C} binding regions could similarly engage PrP^{Sc} without productive conversion, potentially forming a stable, dead-end complex. The following methodology outlines an approach to design and evaluate this construct.

Methodology

Proposed molecular design

36

The construct would use D-amino acids at regions corresponding to PrP^{C}'s key PrP^{Sc}-binding sites (residues 23-33, 98-110, 136-158) [21]. However, substituting the L-amino acids for D-amino acids may result in the side-chain arrangement being significantly dissimilar and unable to bind to PrP^{Sc} therefore these regions would have their sequence reversed to create retro-inverso d-amino acid regions of the original binding regions. This is meant to allow the regions to mimic the side chain arrangement of PrP^{C} [22]. To allow for multivalent engagement, these regions would be connected by an α-helical rich scaffold to mimic the native spatial arrangement of the binding regions in PrP^{C}. Covalent cross-links would be incorporated to enhance structural stability and prevent unfolding. Binding is intended to sequester PrP^{Sc} within a stable, dead-end complex, and thereby reduce availability for interaction with PrP^{C} through the capping of the three regions that would have been critical for binding to PrP^{C}.

D-amino acid substitution confers proteolytic resistance and reduced immunogenicity relative to L-counterparts, extending the construct's functional half-life [23-25]. Disulfide bonds, formed via sulfhydryl groups in cysteine residues, are one candidate for stabilization strategy. Notably, PrP^{C} conversion has been observed to occur without breakage of the intramolecular disulfide bond [26]. However, it is unclear whether this reflects mechanistic necessity or mere coincidence, so disulfide bonds alone may be insufficient to prevent conversion. This motivates the use of zero-length crosslinking as an additional stabilization strategy. Zero-length crosslinking forms a covalent bond between carboxyl and amine groups on different residues of the molecule, which could help in holding the molecule in its normal structure. However, there are challenges to covalent cross-linking. Excessive rigidity may compromise binding to PrP^{Sc} or the formation of a secure dead-end complex, while insufficient rigidity risks conformational rearrangement into a seeding-competent form. An empirical approach is thus proposed: synthesizing a small series of constructs with varying crosslinker spacer lengths and flexibility, followed by comparison of binding affinities, using surface plasmon resonance (SPR), against conversion resistance, using protein misfolding cyclic amplification (PMCA) to identify an optimal rigidity range.

Experimental design

The proposed molecule would be modelled using machine learning models to predict binding outcomes, mapping out efficient synthesis paths, and molecular design. Structure prediction would employ a mirror-image modeling strategy, given that current tools, such as AlphaFold, are trained on L-amino acid data. The construct would first be modeled in its L-enantiomeric form, with resulting coordinates mathematically inverted to yield the true D-structure. Molecular dynamics simulation (using AMBER or CHARMM, both of which support D-residue parameters) would then be run against a similarly mirrored PrP^{Sc} template. If favorable outcomes are predicted, the molecule would be synthesized and evaluated for structural stability and ability to retain the key structural features of the native PrP^{C}, with particular emphasis on the previously identified residues.

The molecule would subsequently be assessed on interaction with PrP^{Sc}. The primary characteristics being investigated include: 1. Ability of the molecule to bind to PrP^{Sc}, through surface plasmon resonance (SPR), which is able to measure binding affinity and real-time kinetics 2. Whether conversion of the molecule occurs through PMCA using the molecule followed by proteinase K digestion and western blotting (the molecule should be labelled with an epitope construct to enable antibody detection), and 3. Stability and persistence of the bond, including the propensity of the molecule to dissociate from PrP^{Sc}, under physiological conditions,

such as in a serum or CSF-mimetic buffer. As PMCA alone may miss partial conversion, circular dichroism would supplement it, confirming retention of α-helical (rather than β-sheet) structure. Real-time quaking-induced conversion should be used to measure if PrP^{Sc} retains its ability for seed conversion after introduction of the D-amino acid construct.

To assess potential interactions with other biologically relevant components, the molecule should also be evaluated in the presence of relevant cellular systems such as microglia cells. This would allow for screening of potential unexpected effects on inflammatory pathways and other cellular responses. The propensity of the molecule for degradation by the body's natural proteases should also be investigated to estimate the length of time the molecule can stay intact in the body. Following these preliminary investigations, appropriate cell-line models should be used to further assess behaviors in intracellular environments. Particularly, in the event that disulfide bonds are used for molecular stability, the reducing conditions present in intracellular environments may compromise disulfide-dependent interactions.

If no significant issues or unexpected effects arise, the next stage is in vivo investigations using mice models. Compared with a control group, primary and secondary outcomes should be investigated. Primary outcomes include changes in the rate of disease progression and toxicity risks. Secondary outcomes include rate of cognitive decline, neuroinflammation, non-specific interactions, and other side-effects. Different doses should also be tested to screen for dose-dependent toxicity and when effects, if any, plateau with further increase in amount of molecule given to the subject. Given that the molecule is likely a large protein-based construct, different drug delivery methods should be considered given the likely destination of the therapeutic in the brain.

Ethical considerations

Prion handling should be minimized as much as possible. Prion strains with limited horizontal transmission and well-characterised incubation periods should be used first during testing. Prion work requires BSL-3 facilities and special decontamination protocols, given that standard sterilization procedures are insufficient.

According to the 3Rs of animal research, the use of animals would be avoided during preliminary investigations. In vivo mouse studies would only be done after in-vitro stages have shown promise, and the number of mice used would be minimal.

29 Ethical approval will be obtained from the following institutional boards: Institutional Biosafety Committee, Institutional Animal Care and Use Committee, Institutional Review Board.

Expected outcomes

1. Binding: D-amino acid construct is expected to demonstrate measurable binding affinity for PrP^{Sc}, comparable to native PrP^{C} to allow effective competition for binding.
2. Conversion resistance: The D-construct is expected to resist PrP^{Sc} templated conformational conversion.
3. Stable sequestration: The resulting PrP^{Sc}-D construct complex is expected to have a low dissociation rate to behave as a dead-end complex.
4. Reduced propagation: Consequently, increasing concentrations of the construct should produce a dose-dependent reduction in PrP^{Sc} seeding/propagation compared with untreated controls.

5. Specificity: D-amino acid construct should mainly interact with PrP^{Sc} and should not have minimal non-specific interactions with other body molecules

Importance

It would provide a proof-of-concept for protein-based prion propagation inhibitors to use enantiomeric mimics of PrP^{C} due to the differences between D-amino acids and L-amino acids and current therapeutics with similar mechanisms use L-amino acid derivatives. It could establish a potential new therapeutic strategy for further optimisation and preclinical investigation. It may also provide broader insight into prion-like diseases and targeting pathological protein-templating mechanisms.

Limitations

The above outcomes assume ideal conditions; several uncertainties remain. The stereochemical tolerance seen in Aβ-PrP interactions may not extend to PrP^{C}-PrP^{Sc} binding. The proposed regions (residues 23-33, 98-110, 136-158) come from prior PrP^{C} literature and are unvalidated for this construct. The dead-end complex may not persist in the long-term, risking PrP^{Sc} dissociation and propagation. D-amino acid substitution may also predispose the construct to non-specific interactions with off-target cellular components. More fundamentally, the molecule likely only halts further propagation; it cannot reverse damage from late-stage diagnosis, when significant neurodegeneration has already occurred. Drug resistance is also possible as PrP^{Sc} strains with lower affinity for the construct may propagate preferentially, passing on resistant conformations.

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